

Insect food for astronauts: gas exchange in silkworms fed on mulberry and lettuce and the nutritional value of these insects for human consumption ...

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In this study, silkworm moth (*Bombyx mori* L.) larvae were regarded as an animal protein source for astronauts in the bioregenerative life support system during long-term deep space exploration in the future. They were fed with mulberry and stem lettuce leaves during the first three instars and the last two instars, respectively. In addition, this kind of environmental approach, which utilised inedible biomass of plants to produce animal protein of high quality, can likewise be applied terrestrially to provide food for people living in extreme environments and/or impoverished agro-ecosystems, such as in polar regions, isolated military bases, ships, submarines, etc. Respiration characteristics of the larvae during development under two main physiological conditions, namely eating and not-eating of leaves, were studied. Nutrient compositions of silkworm powder (SP), ground and freeze-dried silkworms on the 3rd day of the 5th instar larvae, including protein, fat, vitamins, minerals and fatty acids, were measured using international standard methods. Silkworms' respiration rates, measured when larvae were eating mulberry leaves, were higher than those of similar larvae that hadn't eaten such leaves. There was a significant difference between silkworms fed on mulberry leaves and those fed on stem lettuce in the 4th and 5th instars ($P<0.01$). Amounts of CO₂ exhaled by the silkworms under the two physiological regimes differed from each other ($P<0.01$). There was also a significant difference between the amount of O₂ inhaled when the insects were under the two physiological statuses ($P<0.01$). Moreover, silkworms' respiration quotient under the eating regime was larger than when under the not-eating regime. The SP was found to be rich in protein and amino acids in total; 12 essential vitamins, nine minerals and twelve fatty acids were detected. Moreover, 359 kcal could be generated per 100 gram of SP (dry weight).